Brief Report

Palliative Care Physicians' Perceptions about Using Artificial Intelligence for Prognostication

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Abstract

Context. Statistical and artificial intelligence (AI)-based methods have informed clinical prognostication for decades, evolving into machine learning models integrated into electronic health records. Despite increasing deployment of AI-based prognostic tools, palliative care physicians' perceptions of these tools remain understudied.

Objectives. To understand palliative care physician perspectives on clinical use and implementation of AI-based prognostic tools.

Methods. We conducted a national survey of n = 2500 Hospice and Palliative Medicine physicians in the United States (January 2024−March 2025) to assess current prognostic practices, AI knowledge, and perceived benefits and risks of AI-based prognostication.

Results. About 537 completed surveys were included for analysis. Respondents were predominantly White (73.2%) and female (52.9%). Most reported being early technology adopters (69%) with low knowledge of AI (79.3%) and AI-based prognostication (91.8%). Overall, 72.9% routinely provide prognoses; 24.0% have used AI-generated prognoses at least once. Attitudes toward AI were favorable: 70.9% believed AI could facilitate earlier palliative care, 78.0% thought it could improve patients' ability to plan for end-of-life, 62.8% supported its role in hospice eligibility decisions, and 71.7% felt it might reduce team disagreements about prognosis. However, 36.3% were concerned about legal liability, and 37.5% thought it might negatively affect patients' sense of hope. Multivariate analyses found current users were more likely to hold positive beliefs (e.g., AI will help them better meet their patients' needs (aOR: 1.75; CI: 1.09−2.92; $P = 0.026$).

Conclusions. Palliative care physicians report limited current use of AI-based prognostic tools but generally favorable attitudes toward potential benefits, especially among current AI tool users. J Pain Symptom Manage 2026;72:e340−e347. © *2026 American Academy of Hospice and Palliative Medicine. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.*

Key Words

Artificial intelligence, palliative care, prognostication

Key Message

In the first national survey of palliative care physicians about AI-based prognostic tools, use of AI tools was low, but general attitudes towards AI prognostication and anticipated effects of its implementation were positive, especially among those with experience using the tools. However, concerns around tool development and legal liability exist.

Introduction

While some may consider artificial intelligence (AI) generated prognostication models as a relatively recent phenomenon, AI methods were first applied to clinical decision support and prognostic reasoning in the late 1970s and early 1980s, with early expert systems such as MYCIN and related Bayesian inference models.[1,2] In the 1990s, neural networks were applied to ICU

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mortality predictions which evolved into machine learning models through the early 2000s.[3,4] Yet, evidence of widespread implementation of these models into routine clinical practice is sparse. Additionally, even when prognostic models were applied and communicated to clinicians caring for seriously ill patients, such as in the SUPPORT study, key outcomes such as communication about CPR preferences, timing of Do Not Resuscitate orders, and ICU days prior to death were not impacted.[5]

In 2011, the ePrognosis website went live accruing over 500,000 views in the first week.[6] ePrognosis collates multiple prognostic models across unique specific and broad populations to inform a prognostic calculator. However, the site's true impact was related to the widespread dissemination platform as a delivery mechanism to clinicians for practice integration. Robust website tracking metrics support widespread utilization, but we do not have nationally representative data regarding how palliative care clinicians integrate this or other AI-based tools in their practice.

Big changes came in the mid-2020s when AI tools began to be integrated into electronic health records (EHR) to predict mortality in real time. In practice, this has resulted in risk stratification, goals of care or advance care planning prompts, and guiding serious illness communication workflows.[7–9] Prior studies[10–13] have focused on clinician perspectives on AI use including issues related to accuracy, bias, explainability, and potential for harm if AI-derived information is poorly communicated to patients. Despite the growth in AI-driven prognostication, little is known about what palliative care clinicians see as anticipated effects of such tools when implemented in practice. To address this gap, we developed a survey and queried palliative care physicians regarding AI-based prognostication use in end-of-life palliative care.

Methods

A cross-sectional survey design was used to measure perceptions of AI-based prognostication and prediction of mortality among Hospice and Palliative Medicine board certified physicians practicing in the United States. The Colorado Multiple Institutional Review Board deemed this study (21-4902) exempt human subjects research; survey completion implied participant consent.

Survey Development

The initial survey, developed through an iterative process with six research team members, was based on our interdisciplinary expertise in palliative care, ethics, and survey development[14] and an evidence-based review of existing literature regarding physicians' attitudes and beliefs about AI-based prognostication. To our knowledge, this is the first national survey of palliative care physicians regarding AI use. The study's six-member palliative care National Advisory Board, composed of two physicians, a palliative care nurse, a social worker researcher, a patient (who passed away during the study's course) and a family caregiver, reviewed preliminary survey items for general face validity and provided feedback as to how each item addressed the overarching research question about clinician use of AI. We cognitively tested the survey with a convenience sample of five palliative care physicians to explore their understanding of the questions, paying attention to the specific use case of AI (AI-based prognosis information in the EHR). Survey modifications were primarily related to wording of a few items. A formal pilot test of the survey was conducted with 50 palliative care physicians drawn from the sampling frame to ensure effective data collection methods. Pilot data were screened to assess data quality and to determine the prevalence of missing data.

The final 35-item survey (Supplemental Digital Content 1) takes 10−15 minutes to complete. It includes five sections querying physicians about their current prognostic and shared decision-making practices with patients and family caregivers, attitudes and beliefs toward AI, experiences with AI, willingness to use AI-based prognostication, and ethical issues associated with AI. Data on 12 personal and professional demographic characteristics are also collected. Here, we present those items related to the perceived impact of implementation.

Sample

A random sample of 2500 physicians who met inclusion criteria from IQVIA's OneKey healthcare provider database,[15] were surveyed between January 2024 and March 2025. Survey administration occurred in two randomly assigned waves (n = 1250 per wave) to ensure any issues with survey administration could be addressed between waves. We mailed a packet containing a paper survey, a personalized secure link for electronic completion if preferred, and a $50 incentive check for participation. nonresponders were sent a reminder packet at monthly intervals for three months. After the final mailing, remaining nonresponders with valid email addresses (n = 907) were invited to participate by email.

Data Analysis

Surveys were included in the analysis only if they had responses to questions for age, gender, and race/ethnicity (complete-case analysis). These physician characteristic variables, along with self-reported AI prognostication knowledge and current AI user status were hypothesized to be associated with outcomes of interest. Data were entered into R version 4.40.[16]

Self-reported demographic data, clinical practice characteristics, and AI-related attitude outcomes were analyzed using descriptive statistics. We collapsed response categories for six outcomes of interest to create primary outcomes used in modeling. These outcomes were determined by choosing the most important survey items (e.g., one item from each battery) that we hypothesized a priori would be associated with the physician characteristic variables listed above. Associations were tested between physician characteristics and each outcome by constructing a multivariate logistic regression model. Variable construction and outcomes are delineated in Supplemental Digital Content 2.

Results

Of the 2500 sampled physicians, 583 completed the survey yielding a response rate of 32.6%. Since respondents (n = 46) with demographic missing data were removed, 537 surveys were used in the final analysis.

Respondent Characteristics

Personal and professional demographic characteristics are displayed in Table 1. The average respondent was female (52.9%), White (73.2%), had outpatient clinical responsibilities (73.3%, though 54.3% and 48.9% reported community and academic inpatient hospital setting, respectively), and cared for patients with rapid disease trajectories (58.9%). Sixty nine percent of respondents characterized themselves as being an early adopter of technology. More than three-quarters (78.8%) had at least a little knowledge of AI generally (20.7% moderate/a lot) and 60.0% had at least a little knowledge of AI prognostication (8.2% moderate/a lot).

Table 1
Sample Demographic Characteristics

Characteristic	N (%)
Demographic characteristics	
Age	
30–39	148 (27.5%)
40–49	156 (29.1%)
50–59	100 (18.6%)
60–69	90 (16.8%)
70+	43 (8.0%)
Gender	
Female	284 (52.9%)
Male	253 (47.1%)
Race	
Persons of Color (POC)[a]	144 (26.8%)
White	393 (73.2%)
Ethnicity, Hispanic	27 (5.0%)
General AI knowledge	
A little	312 (58.1%)
A moderate amount/a lot	111 (20.7%)
Nothing at all	114 (21.2%)
AI prognostication knowledge	
A little	277 (51.8%)
A moderate amount/a lot	44 (8.2%)
Nothing at all	214 (40.0%)
Unknown	2
Early adopter of technology[b]	
Yes	369 (69.0%)
No	166 (31.0%)
Unknown	2
Clinical practice characteristics	
Medical school appointment (yes)	282 (52.5%)
Disease specialty trajectory[c]	
Rapid	309 (58.9%)
Gradual	86 (16.4%)
Intermittent	91 (17.3%)
Other	39 (7.4%)
Unknown	12
Patient care setting (>1 selection possible)	
Outpatient settings	374 (73.3%)
Unknown	25
NH/SNF	160 (33.0%)
Unknown	50
Hospice facility/program	271 (54.4%)
Unknown	37
Community hospital	271 (54.3%)
Unknown	36
Academic hospital	240 (48.9%)
Unknown	44
Military or VA hospital	34 (7.1%)
Unknown	58

AI = artificial intelligence; NH = nursing home; SNF = skilled nursing facility; VA = veteran's administration.
[a]POC: those identifying as Asian, Black, Middle Eastern, Native American, multi-racial or Hispanic.
[b]Early technology adopter = those who answered "the first," "Not the very first, but I am among the first," and "Earlier than most of my peers" regarding adoption of new technology.
[c]Disease trajectory was created based on the largest share of patients cared for: rapid (cancer—all types and populations), intermittent (cardiovascular disease, pulmonary disease, liver failure), gradual (neurological disease, psychiatric illness, and kidney failure), and other (autoimmune, chronic pain, OBGYN, primary care, pediatrics).

Current Practice Patterns on Prognosis Communication

Table 2 displays data about physicians' current practices related to sharing prognosis. Most physician respondents (65.7%) report being asked always/very often to give mortality predictions, and 72.9% reported providing patients with a prognosis always/very often in practice. The most reported source of prognostic data was clinical trials (77.8%, sometimes/very often/always), but use of tools overall was low: 35.7% of physicians reported using websites sometimes/very often/always. Only 6.5% reported using AI-based tools sometimes/very often/always (though 24.0% reported using AI at least rarely). Regarding disclosure, 47.7% rarely or never tell patients which tool they use, though the vast majority (85.5%) shared with patients their certainty or uncertainty regarding a prognosis. Overall, 17.4% had access to EHR-embedded tools or believed they would within the next year, and of those, a little over one-quarter (26.4%) reported these are or would be AI based.

General Attitudes Toward AI-generated Prognostic Information

We asked physicians about their overall perspectives on AI-generated prognostic information and appropriate use (Table 3). Overall attitudes were positive; for instance, more than three quarters (75.8%) would trust

Table 2
Current Prognostic and Shared Decision-Making Practices

Provide Prognoses In Your Practice, How Often Do You. . .	Always/Very Often	Sometimes	Rarely/Never
. . .get asked by patients, family members, or caregivers about how long they can expect to live?	353 (65.7%)	152 (28.3%)	32 (6.0%)
. . .provide patients with a prognosis regarding how long they can reasonably expect to live?	389 (72.9%)	103 (19.3%)	42 (7.9%)
Tools used			
. . .use data from clinical trials to derive prognosis predictions?	215 (40.0%)	203 (37.8%)	119 (22.2%)
. . .use websites, such as E-prognosis, to derive prognostic information?	61 (11.4%)	130 (24.3%)	344 (64.3%)
. . .use AI-based prognosis predictions?	8 (1.5%)	27 (5.0%)	502 (93.5%)[a]
Tell patients			
. . .tell patients your overall level of certainty/uncertainty regarding your prognosis?	456 (85.5%)	54 (10.1%)	23 (4.3%)
. . .tell patients which tool(s) you used to arrive at your prognosis?	129 (24.1%)	151 (28.2%)	256 (47.7%)
Current and future EHR tools			
	Yes	**No, but plan to implement within the next year**	**No/Don't know**
In your current medical practice do you have access to standardized tools for generating prognostic information, embedded in an electronic health record (EHR)?	74 (13.8%)	19 (3.6%)	444 (82.7%)
Among those who said "Yes" or "No, but plan to. . ."	**Yes**	**No**	**Don't know**
Is this tool based on AI?	24 (26.4%)	38 (41.8%)	29 (31.9%)

[a]About 24.0% reported using AI at least rarely.

AI-generated prognoses sometimes, and nearly 20% would trust it mostly/completely. Moreover, 74.3% would want their own physician to use AI. On the other hand, physicians appeared to have some skepticism about tool development, with 70.7% believing AI systems might be motivated by financial interests. Physicians were also skeptical of EHR-vendor or technology companies developing such tools, with 82.4% preferring development by palliative care colleagues at another institution even over their own. Lastly, when asked about four different use cases of AI-generated prognoses, physicians offered general support (i.e., more than half reported that AI should be used to some extent or a great extent) for hospice eligibility, allocation of scarce resources, and eligibility for preventive services—but were less enthusiastic about using AI for medical necessity determinations by insurers (46.4% reported "not at all").

Anticipated Effects of Implementing AI-generated Prognostic Information

Several items assessed physicians perceived impact of implementing AI-based prognostic information on patients, clinicians, and care teams (Table 4). The majority (65.0%) thought AI could help patients access palliative care who otherwise would not have (i.e., by identifying palliative care needs) to some/great extent; a larger proportion (70.9%) thought AI could help patients access palliative care sooner. Despite anticipated positive effects on patient planning at the end of life (where 78.0% anticipated small/large positive effects), a significant proportion (29.7% and 37.5%, respectively) expressed the potential for negative effects on patients' overall sense of well-being or hope (with only 23.4% and 18.4% anticipating positive effects, respectively).

Physicians expressed optimism about the effects of AI-based prognostication on themselves. Less than 5% anticipated any negative effects on the ability to provide prognostic information to patients, on their own sharpness of prognostication skills, or their abilities to better meet the needs of their patients. However, 36.3% expressed concern about new legal liabilities as a result of AI tools.

Lastly, physicians saw the potential of AI-based prognostic scores to support team-based care. Most palliative care physicians (71.7%) believed it might also help solve disagreements about prognosis among the patient's care team members as they believed care team members frequently (23.5%) and very often (5.8%) disagree about a patient's prognosis.

Multivariate Logistic Regression Results

Statistically significant results from our adjusted multivariate logistic regression for each of the outcomes

Table 3
Physician Perspectives on AI-Generated Prognostic Information and Appropriate Use

General Attitudes Toward Use				
	Great Extent	Some Extent	Very Little	Not at All
To what extent would you want your own doctor to use AI-generated prognostic information in the care you receive?	194 (36.5%)	201 (37.8%)	97 (18.2%)	40 (7.5%)
If your health system, hospital, or employer implemented AI-based prognostication, to what extent do you think doing so would be primarily motivated by financial interests?	51 (9.7%)	321 (61.0%)	121 (23.0%)	33 (6.3%)
	Never	Sometimes	Mostly	Completely
Based on what you know now, how much trust would you place in an AI-generated prognosis?	25 (4.7%)	405 (75.8%)	102 (19.1%)	2 (0.4%)
Perspectives on tool development				
	Your own health system	Your colleagues at another system	A tech company	An EHR vendor
Which of the following groups would you trust most to develop an AI-based prognostic tool?	27 (5.1%)	434 (82.4%)	55 (10.4%)	11 (2.1%)
Appropriateness of use cases				
To what extent do you believe AI prognostic information should be used for. . .	Great extent	Some extent	Very little	Not at all
. . .determinations of medical necessity for coverage decisions by insurers	28 (5.2%)	116 (21.7%)	142 (26.6%)	248 (46.4%)
. . . hospice eligibility decisions	64 (11.9%)	273 (50.9%)	128 (23.9%)	71 (13.2%)
. . . allocation of scarce resources	41 (7.7%)	228 (42.9%)	151 (28.4%)	111 (20.9%)
. . . eligibility for preventive services (e.g., cancer screening, if desired by the patient)	89 (16.7%)	245 (46.0%)	118 (22.1%)	81 (15.2%)

Table 4
Anticipated Effects of Implementing AI-generated Prognostic Information

Effects on Patients					
To what extent do you think using AI generated prognostic information will. . .	**Great extent**	**Some extent**	**Very little**	**Not at all**	
. . .help patients access palliative care who otherwise wouldn't have?	57 (10.8%)	286 (54.2%)	140 (26.5%)	45 (8.5%)	
. . .help patients access palliative care sooner?	81 (15.4%)	292 (55.5%)	122 (23.2%)	31 (5.9%)	
Overall, what effect do you think telling patients their prognosis was generated by a highly accurate AI algorithm would have on their. . .	**Large negative**	**Small negative**	**No effect**	**Small positive**	**Large positive**
. . .ability to plan for the end of life?	9 (1.7%)	24 (4.6%)	83 (15.7%)	293 (55.6%)	118 (22.4%)
. . .overall sense of well-being?	37 (7.0%)	119 (22.7%)	246 (46.9%)	106 (20.2%)	17 (3.2%)
. . .sense of hope?	51 (9.7%)	147 (27.8%)	233 (44.1%)	78 (14.8%)	19 (3.6%)
Effects on clinicians					
Overall, what effect do you think using AI-generated prognostic information would have on your. . .	**Large negative**	**Small negative**	**No effect**	**Small positive**	**Large positive**
. . .ability to provide accurate prognostic information to your patients?	7 (1.3%)	8 (1.5%)	58 (11.0%)	345 (65.5%)	109 (20.7%)
. . .sharpness of prognostication skills?	8 (1.5%)	16 (3.0%)	73 (13.9%)	329 (62.7%)	99 (18.9%)
. . .ability to better meet the needs of your patients?	5 (1.0%)	11 (2.1%)	127 (24.2%)	310 (59.2%)	71 (13.5%)
To what extent do you think using AI-generated prognostic information will. . .	**Great extent**	**Some extent**	**Very little**	**Not at all**	
. . .create legal liabilities for physicians?	32 (6.1%)	159 (30.2%)	248 (47.1%)	87 (16.5%)	
Team-based care					
	Never/Almost never	**Sometimes**	**Frequently**	**Very Often**	
In your experience, how often do care team members disagree about a patient's prognosis?	68 (12.7%)	311 (58.0%)	126 (23.5%)	31 (5.8%)	
	Great extent	**Some extent**	**Very little**	**Not at all**	
Would having access to accurate AI-generated prognostic information help solve disagreements about prognosis among patients' care team members?	38 (7.1%)	345 (64.6%)	131 (24.5%)	20 (3.7%)	

Table 5
Summary of Statistically Significant Results From Adjusted Multivariable Logistic Regression

Outcome	N	aOR	95% CI	*P*-Value
Belief that AI will have a positive effect on ability to meet patients' needs				
Current AI user	522	1.75	1.09, 2.92	0.026
Belief that AI will have a negative effect on patients' sense of hope				
Current AI user	526	0.63	0.41, 0.97	0.039
Would trust AI completely or most of the time				
Current AI user	532	1.93	1.19, 3.07	0.006
Belief that AI will have a positive effect on patients' ability to plan				
Current AI user	525	2.15	1.26, 3.88	0.007
High AI prognostication knowledge	525	0.42	0.21, 0.83	0.012

aOR = adjusted odds ratio; CI = confidence interval.

chosen for modeling are presented in Table 5 (see Supplemental Digital Content 3 for full adjusted and unadjusted results). We found no associations between age, race/ethnicity, or gender and any of the chosen outcomes. Overall, current AI user status was associated with positive attitudes towards AI. Current AI users had higher odds of believing that AI will have a positive effect on their ability to meet the needs of their patients and on patients' ability to plan, and of trusting AI completely or most of the time when compared to nonusers. They also had lower odds of believing that AI will negatively affect patients' sense of hope compared to nonusers. AI prognostication knowledge was only associated with one outcome: those with high self-reported knowledge had lower odds of believing that AI would positively affect patients' ability to plan.

Discussion

In the 1950s, physicians did not routinely share prognostic mortality information with patients, despite reports that patients wanted this kind of information.[17,18] A shift in medical, legal, and ethical standards led to more widespread practice of sharing prognostic information with patients facing life-limiting illness.[19] In our survey, most palliative care physicians report routinely sharing prognosis. Our respondents reported low utilization of AI-generated prognostic information (just 6.5%) and overall saw many potential benefits of AI-generated prognosis including increasing palliative care access, determining hospice eligibility, and aiding in aligning care team differing assessments of prognosis (perhaps seeing AI as a neutral arbiter of accurate prognosis). While respondents report that they do not routinely share from where their prognosis estimates might originate, they reported transparency with their own degrees of certainty or uncertainty in their prognostication.

One of the reasons cited for why physicians withheld prognostic information in the 1950s was out of fear for negatively affecting a patient's well-being or sense of hope. These concerns persist; about a third of our sample worried about the negative effects of AI prognostication in patients' well-being and sense of hope. The tension between truth telling and maintaining hope is an ever-present challenge in care for patients with life-limiting illness.[20,21] Philosophically, the field of palliative care has championed the practice of reframing hope towards symptom control, quality time, and meaning-making and legacy rather than longevity. The implementation of highly accurate AI prognostication models has the potential to catalyze and expediate these hope reframing conversations.

Our respondents expressed little concern regarding the possible impact on their own prognostic skills, a phenomenon known as deskilling. This finding is contrary to some emerging literature,[22] where deskilling has been seen as a major concern of new providers (and a recent study in procedural gastroenterology that showed reliance on AI can develop quickly).[23] It is possible that our respondents underestimate the risk of deskilling; alternatively, recognizing that physicians using clinical judgment routinely overestimate prognosis optimistically by 2–5-fold,[24–26] palliative care physicians may simply see vast potential improvement over the status quo. Future work could explore whether or how prognostication is a core skill in palliative care.

While our study may be the first national survey of palliative care physicians' attitudes towards the use of AI in prognostication, generally, prior studies have demonstrated high acceptability of AI-powered prognostic tools from clinicians, predominantly in the field of oncology.[12,13,27] In previous studies, AI prognostic tools are favorably viewed by clinicians for their higher accuracy, ability for risk stratification, and ability to augment decision support. Despite these overall favorable attitudes towards AI, clinicians have also been cited as the most significant barriers to AI implementation in the clinical setting and, similar to our findings, often express some skepticism related to lack of transparency or explainability, the impact on professional identity, and concerns for workflow disruption.[28–31] Further, clinicians have previously expressed negative attitudes towards peers who are perceived to be heavy AI utilizers.[32] Thus, clinicians' reported attitudes on our survey may or may not translate to real-world clinical adoption. Additionally, clinician attitudes are only one among numerous factors that could influence adoption; other factors include system priorities, cost, feasibility, and patient attitudes.

Results of our multivariate analyses highlight an important finding: current users of AI generally have more positive attitudes related to AI. This is consistent

with previous reports that current users report greater benefits and usefulness, have higher trust in AI system and report lower levels of distrust.[33,34] However, this may not reflect causation; using AI could lead to more positive attitudes, or those with more positive attitudes may be more likely to use AI. As our work is also cross-sectional, we can only report similar associations and cannot determine causation. Very heavy AI users may also be more aware of AI's limitations (bias and hallucinations) as well as threats to privacy and job displacement. This may be reflected in our finding that those with high AI prognostication knowledge are less likely to think that AI prognostic information will help patients plan ahead compared to those with lower AI prognostic knowledge.

Intriguingly, in our study participants preferred AI-based prognostic tools be developed by other institutions, compared to their own institutions, EHR vendors, or technology companies. This finding requires further exploration; while it could be that physicians distrust their own institutions—another item showed concern about financial motivations behind prognostic scores—it could merely reflect limited AI-development capacity at the local level.

In terms of other potential AI downsides, approximately a third of respondents expressed some concerns about legal risks related to AI use which is a particularly difficult balance for physicians. An accepted legal framework that might help inform questions of liability and how it might be shared across clinicians, the health care system, or AI developer/model is lacking. Further, clinicians may find themselves in a difficult conundrum, at legal risk if they have relied too heavily on AI over their own clinical judgment and also at legal risk if they disregard AI informed clinical guidance in favor of their own clinical judgment. This concern may be heightened as AI models become more complex and harder to explain.

Limitations

While we conducted the first national survey of practicing palliative care physicians' perspectives related to AI-based prognostication, our survey bore a low response rate and could exhibit response bias. We performed a nonresponder analysis revealing that female physicians were less likely to respond to our survey, and hospital- or medical center-affiliated addresses had a higher rate of being undeliverable. This may have influenced our findings in unanticipated ways. Additionally, although one-quarter of palliative care physicians responded that they had ever used AI for prognostic assessment at the time of our survey, there has been exponential growth in use of AI tools across health care systems in the past 12−24 months. Given this growth, it is likely that our findings underestimate the current and future penetrance of AI tools within the EHR, and attitudes may be changing rapidly. As we only surveyed physicians, we are unable to understand the attitudes and beliefs of other palliative care team members (advance practice providers, nurses, social workers, spiritual care providers) which may impact the use of AI-based prognostication. Future research could focus on surveying others and comparing these groups to better know their thoughts and feelings about the use of AI-generated prognostication.

Conclusion

Palliative care physicians reported limited implementation of AI-based prognostic tools at this time. While their attitudes were generally favorable towards the use of AI with potential benefits related to increased early palliative care and hospice access, potential downsides related to legal risks were also present. As implementation of AI-derived prognostic models become more widespread in clinical use, further study on palliative care clinician perceptions and impact on practice is warranted.

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Supplementary materials

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References

1. Miller RA, Pople HE, Myers JD. Internist-1, an experimental computer-based diagnostic consultant for general internal medicine. N Engl J Med 1982;307:468–476.

2. Shortliffe EH. Computer-based medical consultations: MYCIN. New York: Elsevier; 1976.

3. Schinkel M, Paranjape K, Panday RN, Skyttberg N, Nanayakkara PW. Clinical applications of artificial intelligence in sepsis: a narrative review. Comput Biol Med 2019;1:103488.

4. Krittanawong C, Virk HUH, Bangalore S, et al. Machine learning prediction in cardiovascular diseases: a meta-analysis. Sci Rep 2020;10:16057.

5. The SUPPORT Principal Investigators. A controlled trial to improve care for seriously ill hospitalized patients. The study to understand prognoses and preferences for outcomes and risks of treatments (SUPPORT) (erratum in: JAMA 1996;275(16):1232). JAMA 1995;274:1591–1598.

6. Yourman LC, Lee SJ, Schonberg MA, et al. Prognostic indices for older adults: a systematic review. JAMA 2012;307(2):182–192.

7. Stamm T, Bader-El-Den M, McNicholas J, Briggs J, Zhao P. Applications of generative artificial intelligence in outcome prediction in intensive care medicine: a scoping review. Front Digit Health 2025;5:1633458.

8. Huang H, Lyu W, Hasan MM, Houser SH. Adoption of machine learning in US hospital electronic health record systems: retrospective observational study. J Med Internet Res 2025;27:e76126.

9. Everson J, Nong P, Richwine C. Uptake of generative ai integrated with electronic health records in US hospitals. JAMA Netw Open 2025;8:e2549463.

10. Hantel A, Walsh TP, Marron JM, et al. Perspectives of oncologists on the ethical implications of using artificial intelligence for cancer care. JAMA Netw Open 2024;7:e244077.

11. Parikh RB, Manz CR, Nelson MN, et al. Clinician perspectives on machine learning prognostic algorithms in the routine care of patients with cancer: a qualitative study. Support Care Cancer 2022;30:4363–4372.

12. Krishnamurthy N, Besculides M, Gorbenko K, et al. Multidisciplinary clinician perceptions on utility of a machine learning tool (ALERT) to predict 6-month mortality and improve end-of-life outcomes for advanced cancer patients. Cancer Med 2025;14:e70137.

13. Hou J, Wang LL. Explainable AI for clinical outcome prediction: a survey of clinician perceptions and preferences. AMIA Jt Summits Transl Sci Proc 2025;2025:215–224.

14. Maul A. Rethinking traditional methods of survey validation. Meas Interdiscip Res 2017;15:51–69.

15. IQVIA. 2026. OneKey healthcare professional database. Available from: https://www.iqvia.com/solutions/commercialization/data-and-information-management/onekey/onekey-hcp-reference-data. Accessed March 24, 2026.

16. R Core Team. R: A language and environment for statistical computing. .R Foundation for Statistical Computing; 2025. Available from: https://www.R-project.org/ Accessed March 24, 2026.

17. Oken D. What to tell cancer patients. A study of medical attitudes. JAMA 1961;1:1120–1128.

18. Kelly WD, Friesen SR. Do cancer patients want to be told? Surgery 1950;27:822–826.

19. Novack DH, Plumer R, Smith RL, et al. Changes in physicians' attitudes toward telling the cancer patient. JAMA 1979;241:897–900.

20. Qama E, Diviani N, Grignoli N, Rubinelli S. Health professionals' view on the role of hope and communication challenges with patients in palliative care: a systematic narrative review. Patient Educ Couns 2022;105:1470–1487.

21. Bigi S, Ganfi V, Borelli E, et al. Perceptions of hope among bereaved caregivers of cancer patients who received early palliative care: a content and lexicographic analysis. Oncologist 2022;27:e168–e175.

22. Wolters Kluwer Health. 2026. Shadow AI: A hidden risk to healthcare. Available from: https://assets.contenthub.wolterskluwer.com/api/public/content/shadow-ai-a-hidden-risk-to-healthcare-pdf?v=9a036220&_gl=1*z8y0hq*_gcl_au*MjEwMTE2NjQzNi4xNzczNzg4NjI5 Accessed March 24, 2026.

23. Budzyń K, Romańczyk M, Kitala D, et al. Endoscopist deskilling risk after exposure to artificial intelligence in colonoscopy: a multicentre, observational study (Erratum in: Lancet Gastroenterol Hepatol. 2025;10(11):e12). Lancet Gastroenterol Hepatol 2025;10:896–903.

24. Christakis NA, Lamont EB. Extent and determinants of error in doctors' prognoses in terminally ill patients: prospective cohort study. BMJ 2000;320:469–472.

25. Glare P, Virik K, Jones M, et al. A systematic review of physicians' survival predictions in terminally ill cancer patients. BMJ 2003;327:195–198.

26. Gwilliam B, Keeley V, Todd C, et al. Prognosticating in patients with advanced cancer: observational study comparing the accuracy of clinicians' and patients' estimates of survival. Ann Oncol 2013;24:482–488.

27. Hildebrand RD, Chang DT, Ewongwoo AN, Ramchandran KJ, Gensheimer MF. Study of patient and physician attitudes toward automated prognostic models for patients with metastatic cancer. JCO Clin Cancer Inform 2023;7:e2300023.

28. Hamer MK, Alasmar AY, Huber K, et al. The ethics of "screening" for end of life using artificial intelligence-based tools: a qualitative study of palliative care team members. J Pall Med 2025;28:1465–1472.

29. Heinrichs H, Kies A, Nagel SK, Kiessling F. Physicians' attitudes toward artificial intelligence in medicine: mixed methods survey and interview study. J Med Internet Res 2025;26:e74187.

30. Scipion CEA, Manchester MA, Federman A, Wang Y, Arias JJ. Barriers to and facilitators of clinician acceptance and use of artificial intelligence in healthcare settings: a scoping review. BMJ Open 2025;15:e092624.

31. Negash S, Gundlack J, Buch C, et al. Physicians' attitudes and acceptance towards artificial intelligence in medical care: a qualitative study in Germany. Front Digit Health 2025;14:1616827.

32. Yang H, Dai T, Mathioudakis N, et al. Peer perceptions of clinicians using generative AI in medical decision-making. NPJ Digit Med 2025;8:530.

33. Horowitz MC, Kahn L, Macdonald J, et al. Adopting AI: how familiarity breeds both trust and contempt. AI Soc 2024;39:1721–1735.

34. Palm R, Kingsland JT, Bolsen T. Trust, experience, and innovation: key factors shaping American attitudes about AI. Public Underst Sci 2026;35:295–312.